

Seeking Principled Diffeomorphic Hylogisms (2/6): Constructing Smooth Quotients from the State-Configuration Space

Pascal Tohour

First Posted: 2026-06-01 (Figshare)

Current Version: 2026-06-16

The author's approach to this 6-part series is to post each part's technical sketch first without proofs, presuming at each step that prior sections hold. Should later scrutiny invalidate prior sections, the author will diminish or amend claims accordingly. The reader should not treat these early iterations as settled.

Abstract

This research note is the second in a series that examines the bridge between the physical universe's symmetric informational descriptions. En-route to a diffeomorphic connection from the state-configuration space to the semantic vector space, this note seeks a functorial hylogism, allowing the principled identification of physical structures across space and scales. The prior note discussed a diffeomorphism from the Borel σ -algebra proxy and mass-configuration space, to the Borel-induced quotient space. This note constructs a synonymous quotient space via the full state-configuration space.

Distinguishability on the State Configuration Space

Having established the ρ – *catalogue* ($\mathcal{E}_\rho$) as the universal domain for mass-configuration identities, a parallel construction is necessary for the richer, complete reality. The hylogism must catalogue every object by mass $\rho(x)$ and all other distinguishing states $\phi_{n_\phi}(x)$ that might inhabit a coordinate. One must therefore separate the mass-density field from other fundamental fields (e.g., the decomposition of mass-density into quark fields or the consideration of other true fields, like the EM and strong nuclear fields [1], [2].) An exact, true field description lies beyond the purview of this note. Instead, the note considers the general setup where an unspecified¹ number of fields occupy a section-indexed space², $\mathbb{S}_{[\phi, \phi]}$.

Make the distinction that $\mathbb{S}$ is both (i) base-space dependent, and (ii) (interpretatively) scale-dependent. Consider base-space dependency. A change in base geometry from 3D to a 4D spacetime engenders the section change ($s \rightarrow \mathcal{S}$) to ($\tilde{s} \rightarrow \tilde{\mathcal{S}}$). So, a section like the extrinsic curvature³ K_{ij} , is no longer canonical when upgrading P to $\mathbb{R}^4$ [3]. More generally, the section space becomes dependent on P (its fields gaining a subscript Φ_P to communicate the P -root.) Consider scale-dependency, modelling choices differ on the sections in $\mathbb{S}$. For example, temperature, a fundamental field in some coarse models, is not fundamental when fine-grained into the mass-density, velocity, stress-tensor, and E-M fields [2], [4], [5].⁴ Articulate the full state tuple in vector notation:

$$S(x) = \begin{bmatrix} \rho(x) \\ \phi_1(x) \\ \vdots \\ \phi_{N_\phi}(x) \end{bmatrix} = \begin{bmatrix} \rho(x) \\ \Phi(x) \end{bmatrix} \quad (1)$$

¹(Possibly infinite in the general case, but this text does not yet prove the extension to arbitrary field cardinality.)

²(or in the infinite case $\mathbb{S}_\infty$.)

³(which specifies how the 3D geometry changes over time.)

⁴Equally, physicists disagree about what the fundamental fields *are* (in the limiting interpretation that fields alone describe nature). Frontier debates about fundamental physical matter lie beyond the ambit of this work's simplified setting.

The bridge through the state-configuration space $\Gamma(P, \mathbb{S})$ differs from the prior mass construction. Let the full state configuration be a global section across P , decomposed into its constituent fields:

$$s \in \Gamma(P, \mathbb{S}), \quad s(x) = (\phi_0(x), \phi_1(x), \dots, \phi_{N_\phi}(x)) \quad (2)$$

where $\phi_0(x) := \rho(x)$. With $|\Phi|$ fields, a single Borel-induced restriction on P is no longer sufficiently permissive; a highly localised mass can induce an infinite-range gravitational field, for example. Therefore, assign a field-specific Borel region $B_{n_\phi} \in \mathcal{B}(P)$ to each component, forming a tuple of observation windows $\mathbf{B} = (B_0, B_1, \dots, B_{N_\phi})$. The local restrictions must apply component-wise:

$$\phi_{n_\phi}|_{B_{n_\phi}} : \Gamma(P, \mathbb{S}_{n_\phi}) \rightarrow \Gamma(B_{n_\phi}, \mathbb{S}_{n_\phi}|_{B_{n_\phi}}) \quad (3)$$

and if the two private universes agree, then one disregards their external differences:

$$s_1 \sim_{\mathbb{S}, B} s_2 \iff s_1|_B = s_2|_B \quad (4)$$

The *component-wise* Borel-induced state quotient is thus [6], [7]:

$$Q_{\mathbb{S}_{n_\phi}, B_{n_\phi}} := \Gamma(P, \mathbb{S}_{n_\phi}) / \sim_{B_{n_\phi}} \quad (5)$$

Consequently, two full physical states are equivalent under a given window configuration $\mathbf{B}$ if and only if they agree across all constituent fields within their respective windows:

$$\mathbf{s} \sim_{\mathbf{B}} \mathbf{s}' \iff \forall n_\phi \in \{0, \dots, N_\phi\}, \phi_{n_\phi} \sim_{B_{n_\phi}} \phi'_{n_\phi} \quad (6)$$

where every pre-image object is an equivalence class:

$$[s]_{\mathbf{B}} = ([\phi_0]_{B_0}, [\phi_1]_{B_1}, \dots, [\phi_{N_\phi}]_{B_{N_\phi}}) \quad (7)$$

The field-wise equivalence condition induces the *total Borel-induced state quotient* as the Cartesian product of the component quotients:

$$Q_{\mathbb{S}_{n_\phi}, B_{n_\phi}} := \prod_{n_\phi=0}^{N_\phi} Q_{n_\phi, B_{n_\phi}} \quad (8)$$

The *disjoint Borel-induced **state** quotient union*⁵ is:

⁵The name is so hefty as to be self-explanatory; a walking definition. Again, also ‘coproduct’, with the full term abridged to ‘s-catalogue’.

$$\mathcal{E}_S = \coprod_{B \in \mathcal{B}(\mathcal{P})} \mathcal{Q}_{\mathbb{S}, B} \quad (9)$$

Just as $\mathcal{E}_\rho$ is an informational index for mass-identity, $\mathcal{E}_S$ serves as the total ‘reality index’. Each quotient space in this co-product is the complete state description (mass, velocity, spin, external fields – whatever one considers the fundamental constituents of reality) occurring in a given window $\mathbf{B}_\Theta$. Invoke an analogous Ehresmann argument for the approximate Borel regions⁶.

$$\mathbf{B}_\Theta = \left(B_{\theta_0, c_0}, B_{\theta_1, c_1}, \dots, B_{\theta_{N_\phi}, c_{N_\phi}} \right), \quad (10)$$

$$\text{where } B_{\theta_{n_\phi}, c_{n_\phi}} = x \in P : \alpha_{\theta_{n_\phi}}(x) \geq c_{n_\phi} \quad (11)$$

In matrix notation:

$$\Theta = \begin{bmatrix} \theta_0 & c_0 \\ \vdots & \vdots \\ \theta_{N_\phi} & c_{N_\phi} \end{bmatrix} \quad (12)$$

with $\Theta \in \mathcal{A}_{reg}^{\mathbb{S}} := \prod_{n_\phi=0}^{N_\phi} \mathcal{A}_{reg, n_\phi}$:

$$Q_\Theta^{\mathbb{S}} := \Gamma(P, \mathbb{S}) / \sim_{\mathbb{S}, \Theta} \quad (13)$$

where

$$s_1 \sim_{\mathbb{S}, \Theta} s_2 \iff \forall n_\phi \in 0, \dots, N_\phi, \quad (14)$$

$$\phi_{n_\phi}|_{B_{\theta_{n_\phi}, c_{n_\phi}}} = \phi'_{n_\phi}|_{B_{\theta_{n_\phi}, c_{n_\phi}}} \quad (15)$$

By the same Ehresmann-type local triviality used in the mass-density case, nearby regular bubbles relate smoothly via diffeomorphisms:

$$\Psi_\Theta : \mathbf{B}_{\Theta_0} \rightarrow \mathbf{B}_\Theta \quad (16)$$

For scalar components, this is ordinary composition. For non-scalar fields, assume a chosen compatible bundle isomorphism over ϕ, θ, c , so that the pullback of sections is well-defined [8]:

$$s|_{\mathbf{B}_\Theta} \longmapsto s|_{\mathbf{B}_{\Theta_0}} \circ \Psi_\Theta \quad (17)$$

⁶Conditional on the properness proof as in the mass-configuration case.

Since this operation acts componentwise on

$$s = (\phi_0, \phi_1, \dots, \phi_{N_\phi}), \quad (18)$$

the nearby quotient fibres $Q_{\theta,c}^{\mathbb{S}}$ locally identify with the reference quotient fibre $Q_{\theta_0,c_0}^{\mathbb{S}}$. Thus, under the prior regularity and properness assumptions,

$$\mathcal{E}_{\mathbb{S}}^{reg} = \coprod_{\Theta \in \mathcal{A}_{reg}^{\mathbb{S}}} Q_{\Theta}^{\mathbb{S}} \quad (19)$$

inherits the same local smooth structure as the mass-density quotient collection. That these statements rely on the same mass-density construction assumptions asserts: (i) that α_{θ} values are regular, (ii) the exclusion of topology-changing bubble events, (iii) the compactness/properness of the underlying domain, and (iv) a chosen smooth structure on the relevant section spaces. If any of these assumptions fail, so too will the smoothness result.

Configuration/Quotient Projections and Categorical Commutativity

Local Projections

Returning to the chair, a full state description filtered to pure mass may preserve the chair's structural identity. One then conjectures that equivalent chair-descriptions in the configuration spaces have a mirrored projection in the quotient space, and if so, whether filtering-then-quotienting is equivalent to quotienting-then-projecting.⁷ Let

$$\pi_{0;|\Phi| \rightarrow 0} : \Gamma(P, \mathbb{S}) \longrightarrow \Gamma(P, \mathbb{S}_0) \quad (20)$$

be the projection that forgets all non-mass state components. A full state section

$$s(x) = (\rho(x), \Phi(x)), \quad (21)$$

filters to

$$\pi_{0;|\Phi| \rightarrow 0}(s) = \rho. \quad (22)$$

Recall (from part 1 and the prior section) that for each Borel observation window $B \subset P$, the quotient maps are:

$$\pi_{\mathbb{S} \rightarrow Q, \mathbf{B}} : \Gamma(P, \mathbb{S}) \longrightarrow Q_{\mathbb{S}, \mathbf{B}}, \quad \pi_{\mathbb{S}_0 \rightarrow Q, \mathbf{B}} : \Gamma(P, \mathbb{S}_0) \longrightarrow Q_{\mathbb{S}_0, \mathbf{B}}, \quad (23)$$

where

$$Q_{\mathbb{S} \rightarrow Q, \mathbf{B}} := \Gamma(P, \mathbb{S}) / \sim_{\mathbb{S}, \mathbf{B}}, \quad Q_{\mathbb{S}_0, \mathbf{B}} := \Gamma(P, \mathbb{S}_0) / \sim_{\mathbb{S}_0, \mathbf{B}}. \quad (24)$$

Recall also the equivalence relations arising from agreement inside $\mathbf{B}$:

⁷Does removing physical detail, then removing external data reveal the same object as first removing data, then detail?

$$s_1 \sim_{\mathbb{S}, \mathbf{B}} s_2 \iff s_1|_{\mathbf{B}} = s_2|_{\mathbf{B}}, \quad (25)$$

and

$$\rho_1 \sim_{\mathbb{S}_0, \mathbf{B}} \rho_2 \iff \rho_1|_{\mathbf{B}} = \rho_2|_{\mathbf{B}}. \quad (26)$$

The projection, $\pi_{0;|\Phi| \rightarrow 0}$, descends to a quotient-level projection

$$\pi_{0;|\Phi| \rightarrow 0, \mathbf{B}} : Q_{\mathbb{S}, \mathbf{B}} \longrightarrow Q_{\mathbb{S}_0, \mathbf{B}} \quad (27)$$

defined by

$$\pi_{0;|\Phi| \rightarrow 0, \mathbf{B}}([s]_{\mathbf{B}}) = [\pi_{0;|\Phi| \rightarrow 0}(s)]_{\mathbf{B}} = [\rho]_{\mathbf{B}}. \quad (28)$$

If two full state configurations agree inside $(\mathbf{B})$, then the states' mass-density components also agree inside $(\mathbf{B})$. Assume that these projections are well-defined. The local physical commutativity square is then:

$$\begin{array}{ccc} \Gamma(P, \mathbb{S}) & \xrightarrow{\pi_{\mathbb{S} \rightarrow Q, \mathbf{B}}} & Q_{\mathbb{S}, \mathbf{B}} \\ \pi_{0;|\Phi| \rightarrow 0} \downarrow & & \downarrow \bar{\pi}_{0;|\Phi| \rightarrow 0, \mathbf{B}} \\ \Gamma(P, \mathbb{S}_0) & \xrightarrow{\pi_{\mathbb{S}_0 \rightarrow Q, \mathbf{B}}} & Q_{\mathbb{S}_0, \mathbf{B}} \end{array} \quad (29)$$

Equivalently,

$$\bar{\pi}_{0;|\Phi| \rightarrow 0, \mathbf{B}} \circ \pi_{\mathbb{S} \rightarrow Q, \mathbf{B}} = \pi_{\mathbb{S}_0 \rightarrow Q, \mathbf{B}} \circ \pi_{0;|\Phi| \rightarrow 0}. \quad (30)$$

Projecting first and then quotienting gives the same mass-identity class as quotienting the full state first and then projecting to its mass-only quotient. So, the *order* in which one removes information from the chair's description does not affect the chair's identity; data-then-detail amounts to detail-then-data.

Global Projection over the Quotient Coproducts

The local quotient spaces assemble into the coproducts

$$\mathcal{E}_{\mathbb{S}} := \coprod_{\mathbf{B} \in \mathcal{B}(P)} Q_{\mathbb{S}, \mathbf{B}}, \quad \mathcal{E}_{\rho} := \coprod_{\mathbf{B} \in \mathcal{B}(P)} Q_{\mathbb{S}_0, \mathbf{B}}. \quad (31)$$

Define the global projection

$$\Pi_{\mathbb{S} \rightarrow \mathbb{S}_0} : \mathcal{E}_{\mathbb{S}} \longrightarrow \mathcal{E}_{\rho} \quad (32)$$

fiber-wise by⁸

$$\Pi_{\mathbb{S} \rightarrow \mathbb{S}_0}(\mathbf{B}, [s]_{\mathbf{B}}) = (\mathbf{B}, [\pi_{0;|\Phi| \rightarrow 0}(s)]_{\mathbf{B}}). \quad (33)$$

Since this mapping exists and is well-defined, the Borel window persists between the co-products. Simply, the mapping allows one to jump from the rich space of all physical bubbles - all objects, into the specific catalogue of mass-forms while preserving the bubbled mass-section. Since mass density is just one filtration of the state space, the spectral-style analysis of filtrations gives an early insight into how quotienting in the final semantic vector space might occur. To include the original configuration spaces in the global statement, define the quotient maps (observer-indexed):

$$\hat{\pi}_{\mathbb{S}} : \mathcal{B}(P) \times \Gamma(P, \mathbb{S}) \longrightarrow \mathcal{E}_{\mathbb{S}}, \quad \hat{\pi}_{\mathbb{S}}(\mathbf{B}, s) = (\mathbf{B}, [s]_{\mathbf{B}}), \quad (34)$$

and

$$\hat{\pi}_{\mathbb{S}_0} : \mathcal{B}(P) \times \Gamma(P, \mathbb{S}_0) \longrightarrow \mathcal{E}_{\rho}, \quad \hat{\pi}_{\mathbb{S}_0}(\mathbf{B}, \rho) = (\mathbf{B}, [\rho]_{\mathbf{B}}). \quad (35)$$

First quotienting to the state co-product and then projecting to the filtered co-product is one feasible route. Another route is to first apply the projection fibre. Then the global physical commutativity relation is

$$\Pi_{\mathbb{S} \rightarrow \mathbb{S}_0} \circ \hat{\pi}_{\mathbb{S}} = \hat{\pi}_{\mathbb{S}_0} \circ (\text{id}_{\mathcal{B}(P)} \times \pi_{0;|\Phi| \rightarrow 0}). \quad (36)$$

In plain terms:

1. choosing a Borel object,
2. quotienting the full state-configuration space,
3. projecting to the ρ -catalogue,

yields the same object as:

1. choosing a Borel object,
2. filtering the full state-configuration space,
3. then quotienting to the ρ -catalogue.

⁸This projection is well-defined because if $[s]_{\mathbf{B}} = [s']_{\mathbf{B}}$, then $s|_{\mathbf{B}} = s'|_{\mathbf{B}}$, and therefore

$$\pi_{0;|\Phi| \rightarrow 0}(s)|_{\mathbf{B}} = \pi_{0;|\Phi| \rightarrow 0}(s')|_{\mathbf{B}}.$$

Hence

$$[\pi_{0;|\Phi| \rightarrow 0}(s)]_{\mathbf{B}} = [\pi_{0;|\Phi| \rightarrow 0}(s')]_{\mathbf{B}}.$$

Diagrammatically:

$$\begin{array}{ccc}
 \mathcal{B}(P) \times \Gamma(P, \mathbb{S}) & \xrightarrow{\hat{\pi}_{\mathbb{S}}} & \mathcal{E}_{\mathbb{S}} \\
 \text{id}_{\mathcal{B}(P)} \times \pi_{0;|\Phi| \rightarrow 0} \downarrow & & \downarrow \Pi_{\mathbb{S} \rightarrow \mathbb{S}_0} \\
 \mathcal{B}(P) \times \Gamma(P, \mathbb{S}_0) & \xrightarrow{\hat{\pi}_{\mathbb{S}_0}} & \mathcal{E}_{\rho}
 \end{array} \tag{37}$$

If the preceding claims endure scrutiny, then this note confirms that a state configuration space and its filtration both have borel-induced quotient co-products, and the global projection allows movement from the unfiltered to the filtered space. These components support the hylogism by allowing objects to map, in an orderly way, to the smooth quotient space, before projection to $\mathbb{V}$.

References

- [1] P. Deligne and D. S. Freed, “Classical field theory,” *Quantum fields and strings: a course for mathematicians*, vol. 1, p. 2, 1999.
- [2] B. Lautrup, “The pde’s of continuum physics,” in *Proceedings of the Workshop on PDE methods in Computer Graphics*, 2005.
- [3] E.ourgoulhon, “3+ 1 formalism and bases of numerical relativity (2007),” *arXiv preprint gr-qc/0703035*, vol. 212,
- [4] K. Robertson, “Asymmetry, abstraction, and autonomy: Justifying coarse-graining in statistical mechanics,” *The British Journal for the Philosophy of Science*, vol. 71, no. 2, pp. 547–579, 2020. DOI: [10.1093/bjps/axy020](https://doi.org/10.1093/bjps/axy020).
- [5] P. Flowers, K. Theopold, R. Langley, and W. R. Robinson, *Chemistry 2e*. Houston, Texas: OpenStax, Feb. 2019, Section 9.5: The Kinetic-Molecular Theory. [Online]. Available: <https://openstax.org/books/chemistry-2e/pages/9-5-the-kinetic-molecular-theory>.
- [6] ProofWiki Contributors, *Universal property of quotient set*, https://proofwiki.org/wiki/Universal_Property_of_Quotient_Set, [Online; accessed 1-June-2026], 2026.
- [7] nLab authors, *Disjoint union*, <https://ncatlab.org/nlab/show/disjoint+union>, [Online; accessed 1-June-2026].
- [8] J. M. Lee, *Introduction to Smooth Manifolds* (Graduate Texts in Mathematics). Springer New York, 2003. DOI: [10.1007/978-0-387-21752-9](https://doi.org/10.1007/978-0-387-21752-9).